

Can AI Work an 8-Hour Day?

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Image credit: @blac.ai

Artificial Intelligence (AI) has been making waves across industries, transforming how we work, think, and innovate. A recent post by DeepFlow on X caught my attention, sparking a deeper dive into the question: “How long until AI can work a full 8-hour day?”

According to the post, some experts predict that by 2027, AI could manage this feat with a 50% success rate. This bold claim is backed by METR’s latest research, which shows AI’s task endurance doubling every 7 months. Systems that currently operate for mere minutes could soon handle hours without human intervention. Let’s unpack this idea and explore what it means for the future of work, technology, and society.

The Exponential Rise of AI Task Endurance

METR, a research group focused on measuring AI performance, has been tracking the length of tasks AI agents can complete autonomously. Their findings are striking: since 2019, AI’s ability to sustain performance on tasks has been doubling roughly every 7 months. This exponential growth mirrors trends we’ve seen in other areas of AI development, such as compute power, data availability, and algorithmic efficiency. For context, a 2023 Time article highlighted how these three pillars—compute, data, and algorithms—have fueled AI’s rapid evolution over the past decade. If METR’s projections hold, by 2027, AI systems could be capable of handling multi-hour tasks with a 50% success rate, a significant leap from today’s capabilities.

To put this into perspective, imagine an AI system that can currently process data or write code for 8 minutes before requiring human oversight. If its endurance doubles every 7 months, by 2027, that same system could potentially work for hours—perhaps even a full 8-hour workday—without needing us to step in.

This isn't just a technical milestone; it's a game-changer for industries that rely on repetitive, modular tasks like data processing, software development, or customer support, or well, any knowledge work really.

The Broader Trends Driving AI's Growth

The exponential growth in AI task endurance doesn't exist in a vacuum. It's part of a larger trend in AI development that has been accelerating for years. As mentioned in the 2023 Time article, the rapid advancement of AI has been driven by three key factors:

1. **Compute Power:** The amount of computational power available for AI training and inference has grown exponentially. For example, a 2025 RAND report notes that AI data centers could require 68 gigawatts of power by 2027—equivalent to the total power capacity of California in 2022. This massive demand underscores the scale of investment in AI infrastructure.
2. **Data Availability:** AI systems thrive on data, and the digital age has provided an abundance of it. However, there's a potential roadblock on the horizon. As tasks become more complex, the data required to train AI for higher-level skills may become exponentially scarcer. A comment in the METR blog post on LessWrong raises this concern, noting that while compute scales exponentially, the data needed for advanced tasks might not keep pace, potentially slowing progress by 2026.
3. **Algorithmic Efficiency:** Advances in algorithms, such as transformer models and reinforcement learning, have made AI systems more capable and efficient. These improvements allow AI to tackle increasingly complex tasks with greater accuracy and endurance.

Together, these factors have created a perfect storm for AI development, enabling the kind of rapid progress METR is observing. But while the trajectory looks promising, there are challenges that could derail this timeline.

The Economic and Social Implications

If AI can truly work an 8-hour day by 2027, the implications for the global economy are profound. A 2016 White House report on automation provides a historical perspective, estimating that automation could displace 15% of the global workforce—around 400 million workers—between 2016 and 2030. However, the same report also projected that new jobs would be created, with labour demand increasing by 21% to 33% over the same period. AI's ability to take on full workdays could accelerate this shift, particularly in industries with clear-cut, modular tasks.

For example, data processing, clerical, and customer service—sectors already seeing significant automation—could be fully transformed.

DeepFlow.com thread highlights that even imperfect AI can be a game-changer economically, as long as the cost of verifying its work remains manageable. However, tasks requiring ambiguity, judgment, and complex dependencies—like strategic decision-making or creative problem-solving—will likely remain a challenge for AI, at least in the near term.

The transition won't be seamless. For example, legacy system integration and verification bottlenecks could slow adoption. If it takes too long to check AI's output, the economic benefits of automation may not materialise as quickly as hoped. Moreover, the societal impact of such widespread automation will require careful management. The 2016 White House report emphasised the need for broad policy responses, including workforce retraining, education reform, and social safety nets, to address the displacement caused by automation. These recommendations are even more relevant today as AI's capabilities continue to expand.

Challenges and Uncertainties Ahead

While the idea of AI working a full 8-hour day is exciting, it's not without its hurdles. The potential scarcity of data for higher-level tasks, as mentioned earlier, could slow progress. If the data required to train AI for complex, judgment-heavy roles grows exponentially harder to obtain, the doubling time for task endurance might lengthen, pushing the 2027 milestone further out.

Finally, there's the human element. DeepFlowAI's thread ends with a thought-provoking question: "What part of your day would stump an AI, even if it had 8 hours to try?" For many of us, tasks involving emotional intelligence, creativity, or ethical decision-making are likely to remain human domains for the foreseeable future, although I think people will increasingly be comfortable with machine playing a significant role here too.

Human-AI collaboration, rather than full automation, may be the more immediate outcome, with AI acting as a powerful sidekick rather than a replacement.

Looking to the Future

The idea of AI working a full 8-hour day by 2027 is both thrilling and daunting. On one hand, it promises unprecedented levels of productivity and efficiency, potentially revolutionising industries and economies. On the other hand, it raises important questions about the future of work, the ethical use of AI, and the societal changes we'll need to navigate.

METR's research gives us a glimpse into what's possible, but the path forward will depend on how we address the technical, economic, and social challenges ahead. As AI continues to evolve, one thing is clear: the future of work will be a partnership between humans and machines, with each bringing their unique strengths to the table.

So, what do you think? Could AI handle your workday by 2027? And more importantly, what tasks do you think will remain uniquely human, no matter how advanced AI becomes? I'd love to hear your thoughts in the comments below.

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Sources:

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